

KAPASA MAKASA UNIVERSITY CAMPUS

COURSE NAME : PRINCIPLES OF CROP SCIENCE AND HORTICULTURE

COURSE CODE : AS 220 and NR 310

IMPORTANT NOTE:

To pass this course better, students are advised to attend contact sessions where detailed information are given. The supplementary notes are summaries of what is covered in contact sessions.

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TOPIC 1: INTRODUCTION

The introduction of agriculture and the gradual movement of humanity into settled communities saw the global population increase gradually to around 300 million by AD 0. During the 20th century, the planet's population tripled in size. The increment in human population was largely due to improvements in **diet**, sanitation and medicine, especially compulsory vaccination against many diseases. Most people agree that population increases will continue, but there are arguments about the rate of increase, and even a few people who believe population decreases are likely. The United Nations has gradually been revising its predictions downwards, and now believes that the world population in 2050 will be around 9.7 billion. It believes that, as the world grows steadily richer and the average family size decreases, growth will steadily slow and eventually stop.

However, others believe that poverty, inequality and continued urbanization will encourage steadily increasing growth, particularly in countries in Africa and parts of Asia, where growth is already much higher than the global average. Others believe that the current world population is unsustainable, and predict that humanity will simply not be able to produce enough food and oil to feed itself and sustain the industrial economy.

1.1 Can growing crops feed the increasing population?

Yes, the current crop production is sufficient to feed the predicted 9.7 billion world population by 2050, but only if we make radical changes to help meet the artificial created shortages. The following are some of the changes;

1. Dietary choices
2. Reducing food wastage;
    - Occurring at all stages of the supply chain.
    - Excess consumption above that required for healthy living.
3. Reconsider food diversions.
4. Work on poor policies
5. Consider the role of agricultural Science in the constitutions of most African countries.

1.2 The world food system

- There has been sustained improvement in food availability since green revolution.
- World food staples crops eaten regularly (even daily) and supply a major proportion of a person's energy and nutritional needs;
    - Cassava

- Maize
- Banana
- Potatoes
- Millet
- Rice
- Sorghum
- Soya beans
- Sweet potatoes
- Wheat
- Yams
- etc.

- The focus in the green revolution era was maize, rice and wheat.
- The problem for the future in the world food system is related to the fact that a solution to the pressure for increased food production when there is population explosion.
- The current estimates of increase in world population shows that there is one birth every 7 seconds and one death every 13 seconds and there is a net gain of one person on earth every 11 seconds.
- Solution to feed world population can be obtained in three ways, namely through:
  - Irrigation
  - Expansion of agricultural land
  - An increase in harvestable yields – Crop science is here!!

1.3 What drives crop food production?

- Production is a function of environment and genotype (Crop Yield = Genotype + Environment).
- All crops (genotypes) have 100% output yields when planted or grown.
- But an environment factor dictates the actual realised yield.
- Below are the environmental factors that influence yields of crops:
  - Physical
  - Biological
  - Chemical
  - Socio-economic environment
  - Infrastructure

- Government policy
- Labour availability
- Technology
- Marketing

1.4 Crop classification

Crops are classified differently;

1.4.1 Based on Use of Crops

Seventy (70%) of the food of the world's population comes from crops. Use of crops can either be:

- **Direct use:**
  - Staple foods
    - The staple foods are the major source of carbohydrates.
  - Food Legumes
    - Food legumes are the major sources of proteins, vegetable oil, minerals, and carbohydrates.
  - Oil crops
    - Oil crops are the major source of vegetable oil.
  - Horticultural crops
    - Horticultural crops are the principle sources of minerals and vitamins.
  - Stimulants
    - Stimulants are the principal sources of caffeine and nicotine.
- **Indirect use:**
  - Alternatively the products of plant growth may be consumed by animals as:
    - Pastures grasses and legumes, cereal grains, roots and tubers
    - Stock feeds to provide food for human beings in the form of meat, eggs, and other animal products
- **Industrial crops:**
  - Products of plant growth can be used as raw materials for industrial processes.
  - Crops that can be converted to alcohols, which can be used for alcoholic beverages and blending into fuels
  - Some plants produce fibre, which can be extracted and used to manufacture textile materials

- Oil from oil seed crops is used in the manufacture of soaps, detergents, plastics, margarine, lubrication oils.
- Transport fuels can be produced from vegetable oils

1.4.2 Based on the Indicative Crop Classification (ICC)

- ICC is consistent with the classification of commodities used in FAO's on-line database, FAOSTAT.
- ICC provides only a broad-level structure for the classification of crops developed based on the Central Product Classification (CPC) (UN, 2004a).
- For Groups 1–8, ICC is consistent with CPC at the group level, and generally consistent at the class level. At the sub-class level, the two classifications are similar.
  - Group 1 is for Cereals
  - Group 2 is for Vegetables and melons
    - Leafy or stem vegetables
    - Fruit-bearing vegetables
    - Root, bulb, or tuberous vegetables
    - Mushrooms and truffles
    - Vegetables
  - Group 3 is for Fruit and nuts
    - Tropical and subtropical fruits
    - Citrus fruits
    - Grapes
    - Berries
    - Pome fruits and stone fruits
    - Nuts
    - Other fruits
  - Group 4 is for Oilseed crops
    - Soya beans
    - Groundnuts
    - Other temporary oilseed crops
    - Permanent oilseed crops
  - Group 5 is for Root/tuber crops with high starch or inulin content
  - Group 6 is for Beverage and spice crops

        - Beverage crops
        - Spice crops
        - Permanent spice crops
    - Group 7 is for Leguminous crops
    - Group 8 is for Sugar crops
    - Group 9 is for Other crops
        - Grasses and other fodder crops
        - Fibre crops
        - Medicinal, aromatic, pesticidal, or similar crops
        - Rubber
        - Flower crops
        - Tobacco
        - Other crops

1.4.3 Based on Scientific Classification

- Botanical or scientific names of plants which consist of genus and species and are universally accepted.
- Carolus Linnaeus, a Swedish botanist, was responsible for the binomial system of classification.
- Scientific names derived primarily from Latin, with some derived from Greek.
- Maize = Zea mays L. and Soya beans – Glycine max L.
- In scientific classification, a plant is given one name-its scientific, technical, or botanical mime.
- According to the internationally accepted rule, each plant has a two-word or binomial name given in Latin.
- The first name refers to the genus and the second to the species.
- The initial of the person (authority) who names a plant species or variety is listed after the species name. For example, in the name Triticum aestivum L., 'L' means Linnaeus, who named the wheat plant.
- The generic name always begins with a capital letter and the species name with a lower-case letter.
- These names are underlined if written by hand or typewriter and italicized when printed.
- The name or abbreviated name of the authority appears in Roman (not italic) type.

TOPIC 2: CROP YIELD POTENTIAL

Yield potential is defined as the yield of a cultivar (mainly regulated by crop genetic characteristics) when grown in environments to which it is adapted, with nutrients and water non-limiting; with pests, diseases, weeds, lodging, and other stresses effectively controlled (ideal conditions). As such, it is distinguished from potential yield, which we define here as the maximum yield (actual yield) which could be reached by crop in given environments, as determined, for example, by simulation models with plausible physiological and agronomic assumptions. In most cases, a crop's yield potential is not shown for plants that have been grown using rain-fed. It also means that crop yield potentials are higher genetically than what we get in fields (actual yield). Several implications of the definitions given above are considered, particularly those arising from cultivar interactions with agronomic practices and with the biotic and abiotic environments.

Some scientist believes that with time crop yields potential have been improving; now it has reached a plateau (you can no longer increase yields through breeding and or manipulation of environment). This is the case of developed countries. Others still believe that there are still small improvements in yield (gradual increase can be seen). This is the case of developing countries. With two diverged views on crop yields, two different strategies are required.

- Plateau
    - To shift the plateau higher in order to provide food for ever increasing population.
- Gradual increase
    - To strive to reach the plateau.

2.1 Can we improve crop yields?

- Yes, through breeding programmes.
- Use of fertilisers
- Use of disease and pest control strategies
- Weed control through use of herbicides
- Use of growth regulators
- Through protected agriculture where food crops are produced continuously irrespective of the time of the year
- Mechanisation of field operations
- Use of certified seed only
    - No traditional crops, due to unavailability of certification system.
    - Handling is controlled to ensure good germination.

- Use of biotechnology
    - Tissue culture/cell culture which depends on totipotency.
    - Genetic engineering or genetically modified organisms (GMOs) where crops are purposefully manipulated in order to produce a determined outcome fitting any environment.

2.1.1 Advantages of using biotechnology

- Environmental protection.
- Resistance to diseases and insect damage
- Sustainability
- Increased flavour and nutrition
- Increased storability
- Affordability or decreased food prices
- More income to producers
- Efficient and convenient way to keep germplasm
- Environmental fit crops

2.1.2 Disadvantages of using biotechnology

- Cross pollination challenges
- Rise in allergic reactions
- Not 100% environmentally friendly
- Decreased antibiotic efficacy
- Unusual taste
- Not totally safe to eat
- Conflicts
- Creation of new diseases or resistance breakdown
- Food supply and plant seed sales at risk
- Cost of equipment and need for specialised skills
- Ethical consideration

2.2 Various disciplines crop science agronomists depend on to produce food crops

Even as we are able to improve crop yield potential we must all understand that crop production can be achieved through combined efforts from various disciplines;

- Soil science
- Climate and weather (Agricultural Meteorology)
- Plant breeding or Genetics
- Plant physiology
- Plant pathology
- Plant pest sciences or Entomology
- Weed science
- Management
- Economics
- Marketing
- Legislation

2.3 Plant factors affecting crop yields

We need to understand plant physiology before we can shift the yield plateau or keeping increasing the gradual yield increase to higher levels. Yield comes from plant dry matter (DM), and DM production is a function of photosynthesis. Photosynthesis depends on factors such as water, sunlight, carbon dioxide and nutrition. All other factors (environmental) can be controlled or manipulated by man but not light. However, light landing on plants can be improved to get the best yields (Light use efficiency). If this is true, photosynthesis can be manipulated to increase on DM production.

2.3.1 The Process Photosynthesis

Note: Photosynthesis will be covered through an assignment

- Photosynthesis can occur between light wave length of 250 – 700 nm, but photosynthetic active radiation (PAR) is between 400 and 700 nm
- CO_2 + H_2O =>Glucose + Oxygen + H_2O
    - Pigments that absorb light in green plants (Beta carotene, Chlorophyll a, Chlorophyll b, Phycoerythrin, Phycocyanin)
    - Chloroplasts are the sites of photosynthesis
    - While any green part of a plant has chloroplasts, the leaves are the major site of photosynthesis for most plants.
    - There are about half a million chloroplasts per square millimetre of leaf surface.

- The colour of a leaf comes from chlorophyll, the green pigment in the chloroplasts.

- Glucose is used by the plant for growth and development through assimilation of processes to form products harvested for food, fibre or any product.
  - Glucose can be used to make other sugars, proteins, fats and oils, cellulose, vitamins, organic compounds and lignin

2.3.2 Photosynthetic efficiency

- This is crop's ability to convert solar energy into chemical energy and it is indirectly measured as DM.
  - Solar energy (100%) => Dry matter (< or = 1%)
- Crop producers must know the photosynthetic efficiency at which their crop is operating.
- Improving photosynthetic efficiency is critical if we are to improve on yield (DM)
  - Under low efficiency, improvements can be made
  - Under high efficiency, you need to know the upper limit and see if improvements can be done
- Any light reflected by plants is not used for photosynthesis but lost completely
- Photorespiration and respiration reduces the energy efficiency because it consumes the produced DM
- Economic yields such as grain, oil content or digestible dry matter are often used as common measures of the success of crop production rather than photosynthetic efficiency.

2.3.3 Calculating the photosynthetic efficiency

- Based on
  - Geographical location or area which determines the solar energy received per hectare of crop.
  - Crop growing period (crop duration in the field)
  - Energy required to produce a kg of glucose (15,792 kilojoules)
  - Assumption of DM used by plant for respiration (life maintained) always put at 25%.
  - You need to know that plants cannot grow without mineral nutrients which are part of DM. it is estimated at 10% of total DM as ash.
  - Economic yield harvested at specific moisture content
  - Total DM is given as economic yield + stover + roots (dry weight of roots and stover)
- Example: Suppose a maize crop in Chinsali grown for 100 days produced 6, 200 kg ha^{-1} when harvested at 15%MC. Stover DM was 6,270 kg ha^{-1} and roots DM 4,480 kg ha^{-1} and suppose the solar energy received was 20,953 million KJ ha^{-1}, calculate photosynthetic efficiency of the maize crop.

Solution

- PE = 1.474216 which is approx. 1.5%
- 100% energy fall on green leaves of maize and only 1.5% was captured and carried through DM.
- Thus, we can breed for higher photosynthetic efficiency which ranges from 1 – 6% and improve on yield

2.3.5 Can we improve on photosynthetic efficiency?

1. Leaf area index (LAI) = This is the one-sided area of green leaf tissue per plant unit area of land occupied by that crop. Leaf area and its distribution over land area is one of the major factors that determine light interception, which affects photosynthesis, transpiration and dry matter accumulation.
2. Increasing plant density increases LAI and in turn increases yield performance because you are increasing light capturing (reduces unused light loss). Yield on individual plant decreases while for the overall population it increases with increase in population (draw the graph). LAI increases with increase in plant population to optimum and then declines. Critical LAI is the LAI that results in 95% light interception (maximum). Narrow spacing forms full canopy early and enhances light interception (due to increase in number of leaves capturing light. However, care is taken to avoid some leaves becoming parasites.
3. Ratio of top to roots plays a major role. Depending on the economic part, the ratio plays a major on the utilisation of DM and puts pressure on photosynthesis. For root crops a lower ratio is preferable so that more DM goes to roots while for leaf and grain crops a higher ratio is important.
4. Choice of cultivars (crop species) with inherent efficiency to utilise energy than others and will have a high rate of photosynthesis.
    - The yield of soya bean tend to be 1/3 that of maize. What is the explanation?

TOPIC 3: CROP GROWTH PHASES

3.1 Establishment phase

- Establishment begins with germination and leads to emergence of seedlings.
- Characterized by a slow increase in crop growth indicators (plant height, leaf size and canopy formation) due to the low LAI resulting in a low solar radiation interception.
- Basically the establishment of a plant mostly relies on viability and germination capacity of seed.
- It is a phase of growth characterized by a slow rate of growth where the newly emerged seedlings interact with a new habitat.
- The phase usually takes 21 day period from seeding/planting maize and depends on cultivar.
- On average the phase takes 15 - 25 days.
- The slight difference in establishment phase might be attributed to variations in viability of seeds, germination capacity of seed, soil temperature and moisture.

3.2 Vegetative phase

- The phase follows establishment
- There is a subsequent increase in photosynthesis.
- In essence, the vegetative phase represents the period between establishment and the beginning of sexual maturity.
- During this growth phase, a plant will be photosynthesizing as much as possible to grow as large as it can, before the onset of flowering (reproductive phase).
- The rate of maize growth between plant emergence and tassel emergence most significantly affects the total time required to maturity and establishes the date it will be ready for harvest.
- The phase takes 25-40 days after establishment and is cultivar dependent in maize.
- For instance, this period is shorter for an early maturing hybrid (70 – 80 days) than a medium (100 – 120 days) or late maturity (120 – 160 days).
- In annual crops, the vegetative phase is generally terminated by the onset of flowering where leaves, stems and other vegetative parts fail to compete for current assimilate as grain filling requires a reserve as the main sink.

3.3 Reproductive phase

- The period from silking to physiological maturity is uniform and averages from 50 to 55 days for most hybrids.

- Pollination generally occurs within one to three days after silking and sufficient soil moisture levels, and optimum temperatures are critical for pollination.
- Cob growth also accelerates during this period with the onset of grain filling.
- Approximately 15 days after silking, depending on the maize cultivar, leaf and stalk growth is terminated and sugars produced by photosynthesis in the leaves move into the grain where they are converted into starches, protein and oils.

3.4 Maturation phase

- Grain development is rapid during the next 30 to 35 days.
- Dry matter accumulation in the maize kernels begins shortly after fertilization and progresses in a sigmoid pattern in which three phases can be distinguished.
- **The first phase** corresponds to the lag phase, which is a formative period during which sink capacity is set. It is characterized by a rapid increase in kernel water content with little dry matter deposition.
- **The second phase** of seed growth is known as the effective grain filling period and involves active biomass accumulation, which is generally more important than the lag phase in actual size determination. During this phase, kernel water content reaches its maximum and begins to decline, closely coordinated with dry matter deposition.
- **In the third phase**, kernels achieve their maximum dry weight (commonly referred to as physiological maturity) and enter a quiescent state.
- Variation in final grain weight reflects the interaction between source capacity and sink strength (i.e., the source/sink ratio) during the effective grain filling period.
- Maize grain yield is mainly determined by kernel size and number per unit land area.
- These grain yield components are positively related to crop growth around silking, and biomass allocation to the cobs.

TOPIC 4: ENVIRONMENTAL STRESS AND YIELD

Stress is defined as a condition(s) which affect normal plant growth and development. Such a condition can be in a form of sub-optimal and excessive state of a particular condition. Stress that occurs during the reproductive phase is more fatal than at maturation phase. The reason is that during the time of flowering the number of fruits and grains are determined such that any stress that affects the crop cannot be reversed.

- Pests and disease attacks due to insufficiency of minerals is due to sub-optimal conditions
- Too much of nitrogen is an excessive condition and that plants will keep on going

Specific stress conditions

4.1 Plant population stress

- Stress due to plant population arises from the differences in number of plants per unit area of land with respect to amount of available resources that the plant can utilise (plant density or population).
- Resources (water, light, CO_2 and nutrients) utilisation increases with plant density increase till optimum plant density is reached.
- Optimum plant population is that which gives optimum yield under normal conditions.
- However, during over population, yield in cereals are moderated by compensation.
- Planting less seeds causes lower population
- Over planting leads to over population which causes
    - **Lodging**
        - Defined as the falling of plants due to (i) weak stems or (ii) poor root development.
        - Lodging simply means poor structural components in the plant such as lignin, cellulose and hemicellulose which forms during the secondary growth.
        - There are various causes of lodging in addition to poor structural component formation.
    - **Barrenness/sterility**
        - Defined as the inability to bear an offspring.
        - This is always due to reduced development of female flower as a result of reduction in supply of photosynthates during the reproductive phase.
        - Delayed flower development means no synchronisation with that of male flower.
        - If the two flower formations are off season, no fertilisation will occur.

        - The mature pollen will land on the undeveloped stigma and will fail to germinate and develop.
        - Pollen has a short life span; it dries faster and has shorter life span than the ovule.
    - **Plant death**
        - As the number of plants per unit area of land increases also a high number of plant death increases.

4.2 Nutrient stress (Soil fertility stress)

- Several nutrients cause stress.
- Especially when they are in low supply or excess supply.
- For more details, read topic 7.

4.3 Disease and pest stress

- Severe disease and pest attack results in in poor plant growth ultimately reducing yield.
- Pests also act as hosts for some diseases.

4.4 Interspecific competition stress

This is stress from different competition in a crop growing; natural vegetation, weeds and mixing copping.
Mixed cropping

- Beneficial effects
- Cooperation between crop mixers can lead to higher yields when the mixture has more yield than the highest yield of the two crops (A + B > A). Example maize and groundnuts especially when leaves of maze are erect to allow light to strike the ground nut leaves; Maize and cotton, the two crops extract minerals at different root depth
- Cooperation of crops with similar needs results in competition.
    - The competition will depend on the individual needs.
    - Use of Land Equivalent Ratio (LER) gives a guide how the two crops can grow in a mixture for us to obtain higher yields.
    - When LER is 1, no advantage of growing the crops as a mixture, less than 1 stop the mixture but we need more than 1

4.5 Soil moisture stress

It results from:

- Deficient of water supply in root zone. Deficient is due to reduced rainfall or reduced irrigation.
- Due to excessive atmospheric demands of water from the leaves through transpiration.
- Transpiration is enhanced by high temperature and low humidity.

4.5.1 Importance of transpiration

- Movement of nutrients
- Cooling of plant canopy
- Maintenance of turgor pressure
- Movement of cell solutes (sugars and sinks to sinks)

Transpiration coefficient (WUE) is the amount of water in litres required to produce 1 kg of DM. It depends on

- Soil moisture conditions
- Crop nutrition
- Climate
- Genotype or species

CROP SPECIES	WUE
Sorghum	277
Maize	349
Sugar beet	443
Spring wheat	491
Barley	527
Potatoes	575
Oats	583
Rye	634
Clover	783
Lucerne	844

4.5.2 Plant processes affected by water stress

- Cell growth
- Cell wall synthesis
- Protein synthesis
- Stomatal opening
- Carbon dioxide assimilation
- Nitrate reductase enzyme
- Respiration
- Sugars, proline and abscisic acid accumulation

4.5.3 Plant recovery from water stress

- Depends on
    - Stress severity
    - Crop growth stage when stress occurs
- In case of maize, the following will be affected on plant growth;

  Vegetative
    - Depending on severity, it has less impact
    - Stems, leaves and roots affecting photosynthesis

  Reproductive/flowering
    - Pollen development, pollination, silking process and fertilization
    - Cobs per plant, number of grains per cob, size of cobs
    - No yield component compensation occurs

  Maturity
    - Main activity is the filling of economic portion with produced soluble sugars
    - Stress reduces supply to harvestable parts (grains, tubers, fruits, etc.) and reduced biosynthesis of proteins, lipids and starch.
    - Stored food reserve remobilization for filling of economic sinks can occur to allow some yields but will be low yields

4.5.4 Alleviation of water stress

- Use of anti-transpirant
- Increase the amount of light reflection.

TOPIC 5: CROP PRODUCTIVITY AND YIELD COMPONENTS

Crop yield is affected by physiological factors such as;

- Plant architecture:
    - Refers to the way the leaves are on the plant are spread or
    - The tillering habit of the plant, plant height, number of leaves, size of leaves (L x B)
    - Leaf architecture refers to leaf orientation and their sizes
- Therefore in a day to day context, plant architecture refers to plant morphology.
- Implication of plant architecture on crop production
    - Influences on some agronomic aspects of crop production.
    - This leads to more DM being produced for growth and development of the crop.
- K- value (extinction coefficient)
    - Defined as the measure of effectiveness of obstruction of light penetration.
      Or the degree of shading in the in the plant canopy.
    - Crops with low K-value are tolerant to high leaf area index.
    - (Yield will increase to optimum LAI and then decrease). Similarly, light interception will increase with LAI till critical LAI and then plateau.
    - BEER – LAMBERT's LAW
      His law state that the rate of photosynthesis depends on transmitted light in the canopy which is related to LAI through K-value

5.1 Plant Canopy photosynthesis

- Defined as the photosynthesis in the field situation when plants are growing.
- This is appreciated by growing understanding on leaf photosynthesis.
- Leaf photosynthesis increases with increased light interception or energy.
- The maximum CO_2 the leaf can fix is attained at much lower light radiation (energy).
- Light saturation is attained at much lower level of radiation with leaf photosynthesis.
- Whole canopy photosynthesis continues up to much higher levels of radiation.
    - Leaf morphology refers to thickness and leaf size (narrow vs broad or wider vs thinner).
    - Manipulating leaf morphology can increase light penetration that will lead to higher crop yield.

5.2 Yield and yield Components

5.2.1 Types of yield

- Biological
    - Whole DM (shoots and roots) at 0% MC
    - Most cases, roots are not included due to difficulties.
- Economic
    - Only the harvestable yield is considered
    - Grains, roots/tubers and stems, etc.
    - Quality depends on size of tubers, roots, grains and colour after ripening.
- Harvest index is the proportion of economic yield to biological yield. It gives an idea of DM partitioning.
    - Suppose two maize varieties with harvest index A = 0.42 and B = 0.50, then variety B is better (good in yield)
    - However always refer to conditions under which the crop was produced when reporting about harvest index.

5.2.2 Yield components

- Plants always compensate for reduction in photosynthesis using their yield components.
- Depends on plant species;
    - Small grains
        - Tillers
        - Spikelets
        - Number of grains
        - Mass of grains
        - Plant population
        - etc
    - Big grains
        - Plant population
        - Number of cobs
        - Size of cobs
        - Grains per cob
        - Mass of grains

        - etc
    - Stems and leafy crop
        - Size of stem or leaf
    - Fruit crops (tomatoes)
        - Number of trusses per plant
        - Flower number per truss
        - Number of fruits per truss
        - Number of fruits per plant
        - Mass of fruits per truss
        - Mass of fruits per plant
        - Individual fruit mass
        - Number of primary branches per plant
        - Plant height
- Yield is determined by its yield components
- In case of maize; Yield = number of plants per hectare x number of cobs per hectare x number of grains per cob x mass of grain.
- All yields components are affected by stress at specific time of its occurrence and compensation only occurs in the subsequent yield component formed in the next growth stage.

TOPIC 6: SOURCE –SINK RELATIONSHIP

6.1 Source

- This is the photosynthetic capacity to provide assimilates (leaves and other photosynthetic areas of a plant).
- Husks, leaf sheath and awns all contribute to photosynthesis.

6.2 Sink

- This is the storage capacity of economic yield component.
- Other parts of plant especially those which do not fix CO_2. Even leaves become sinks when they are shaded and before they reach 2/3 of their normal size.
- Roots constitute one of the biggest sinks in plants.
- Sinks differ in accordance to species.
    - Some by nature are stronger than others (Maize cob is stronger than common beans pod and cassava root is stronger that maize cob).
    - Common beans require sugars for Nitrogen fixation, which reduces the size of sinks and their numbers i.e. more pods.
    - Maize cob is more central and one cob maximum two or three thus stronger compared to the common beans with several pods. Yet, maize cob receives assimilates from both halves of the maize plant.
    - Cassava root is the only sink being fed by the phloem.

6.3 Assimilate transport

- The source will primarily export its products to the nearest sink during the movements of assimilates in the plant. The bigger the size of sink, the higher the demand for assimilates.
- Translocation of assimilates comes about after the source has satisfied its maintenance requirements (Maintenance + Growth)
- Translocation of assimilates is influenced by feedback from the sink.
- The rate at which translocation occurs depends on capacity of sinks; tuber size, grain size.
- When no more assimilate are able to be translocated to sinks, rate of assimilate production is slowed. The source stores it and when more storage has occurred, CO_2 fixation is stopped.
- Example: sugar cane and sweet sorghum => bottom portions are always sweet due to deposition of sugars compared to top portions. Secondly, sugars on top portion are continuously for energy production and assimilate production

The movement of organic food materials or the solutes in soluble form from one place to another in higher plants is called as **translocation of organic solutes**.

Translocation of organic solutes is essential in higher plants because:-

- In higher plants, only the green parts can manufacture food and it must be supplied to other non-green parts for consumption and also for storage.
- During the germination of the seeds, the insoluble reserve food material of the seed is converted into soluble form and is supplied to the growing regions of young seedling till it has developed its own photosynthetic system i.e., leaves.

Translocation of organic solutes always takes place from the region of higher concentration of soluble form i.e., the **supply end (source)** to the region of lower concentration of its **soluble form** i.e., the **consumption end (sink).**

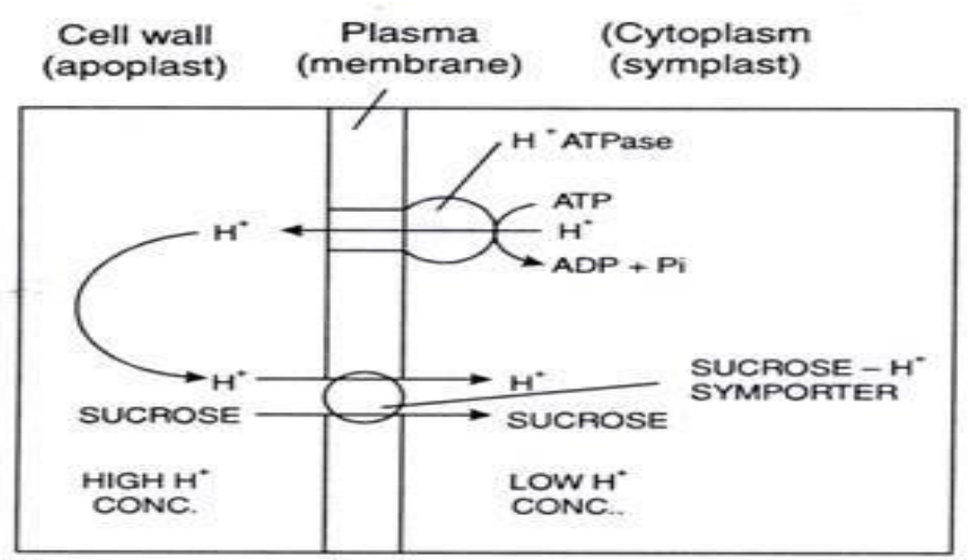
6.3.1 Directions of Translocation

- **Downward Translocation:** Mostly, the organic food material is manufactured by leaves and is translocated downward to stem and the roots for consumption and storage.
- **Upward Translocation:** It takes place mainly during the germination of seeds, tubers etc. when stored food after being converted into soluble form is supplied to the upper growing parts of the young seeding till it has developed green leaves. Upward translocation of solutes also takes place through stem:
    - To buds which resume growth in the spring
    - To developing leaves situated closer to its apex
    - To opening flowers and developing fruits which are situated near the ends of the branches.
- **Radial Translocation:** Radial translocation of organic solutes also takes place in plants from the cells of the pith to cortex.

6.3.2 Phloem Loading

- The transfer of sugars (photosynthetic) from mesophyll cells to sieve tube elements in the leaf is called as **phloem loading**. Energy is required for the process.
- As a result of photosynthesis, the sugars such as sucrose produced in mesophyll cells move to the **sieve tubes of smallest veins** of the leaf either directly or through only **2 - 3 cells** depending upon the leaf anatomy.
- Consequently, the concentration of sugars **increases in sieve tubes** in comparison to the surrounding mesophyll cells.

- The movement of sugars from mesophyll cells to sieve tubes of phloem may occur either through **symplast** (i.e., cell to cell through plasmodesmata, remaining in the cytoplasm) or the sugars may enter the **apoplast** (i.e., cell walls outside the protoplasts) at some point enroute to phloem sieve tubes.
- In the latter case, the sugars are actively loaded from apoplast to sieve tubes by an **energy driven transport** located in the plasma membrane of these cells. The mechanism of phloem loading in such case has been called as **sucrose-H^+ symport** or **cotransport mechanism.** According to this mechanism (Figure below) protons (H^+) are pumped out through the plasma membrane using the energy from ATP and an ATPase carrier enzyme, so that concentration of H^+ becomes higher outside (in the apoplast) than inside the cell. Spontaneous tendency toward equilibrium causes protons to diffuse back into the cytoplasm through plasma membrane coupled with transport of sucrose from apoplast to cytoplasm through **sucrose -H^+ symporter** located in the plasmamembrane



- Experimental findings have revealed certain patterns in apoplastic and symplastic loading of sugars in phloem (see table below), which appears to be related with the type of sugar transported to phloem, type of companion cells (ordinary, transfer or intermediary) and number of plasmodesmata (few or abundant) connecting the sieve tubes (including the companion cells) to surrounding cells in smaller veins.

	Apoplastic loading	Symplastic loading
Type of sugar transported	Sucrose	Sucrose + other oligosaccharides
Type of companion cells in the small veins	Ordinary or transfer cells	Intermediary cells
Number of plasmodesmata connecting the sieve tubes (including companion cells) to surrounding cells	Fewer	Abundant

- To some extent, phloem loading is also correlated with the family of plant, its habit (trees, shrubs, vines or herbs) and climate such as temperate, tropical or arid climate.

6.3.3 Phloem Unloading

- The transfer of sugars (photosynthetic) from sieve tube elements to the receiver cells of consumption end (i.e., sink organs) is called as **phloem unloading**.
- It occurs in the consumption end or sinks organs (such as developing roots, tubers, reproductive structures etc.)
- Steps in movement of sugars from sieve tubes to receiver cells in the sink :
    - **Step 1** is sieve element unloading: In this process, sugars (imported from the source) leave sieve elements of sink tissues.
    - **Step 2** is short distance transport: The sugars are now transported to cells in sink by a short distance pathway which has also been called as post-sieve element transport.
    - **Step 3** is storage and metabolism: Finally, sugars are stored or metabolized in the cells of the sink.
- As with the phloem loading process, sucrose unloading also occurs through symplast via plasmodesmata or through apoplast at some point enroute to sink cells.
- Phloem unloading is **typically symplastic** in growing and respiring sinks such as meristems roots, and young leaves etc. in which sucrose can be rapidly metabolized. (Young leaves act as sink until their photosynthetic machinery is fully developed, at which point they become sources).
- Usually, in storage organs such as fruits (grape, orange etc.), roots (casava, potatoes, sweet potatoes) and stems (sugarcane), sucrose unloading is known to occur through apoplast. However, phloem unloading in potato tubers from sieve elements to cortical cells is a symplastic passive process. Because, there are wide varieties of sinks in plants which differ in structure and function, no one scheme of phloem unloading is available.

6.3.4 Assimilates Partitioning

- The products of carbon assimilation or photosynthesis such as hexoses, glucose, sucrose, starch, fructose, etc. are called as photosynthates or photoassimilates or simply as assimilates. These assimilates are produced in green leaves of higher plants which constitute the sources. Within various compartments of photosynthesising cells (sources), these assimilates are;
    - Metabolically utilised
    - Stored
    - Converted into transport sugars mainly sucrose for export to various sinks (through phloem) such as young leaves, roots, tubers, stems, fruits and seeds.
- Although attempts to increase photosynthetic activities of the leaves have met with only very little success, but the harvest index (ratio of the harvest yield such as grains in beans to total shoot yield) or yield of many crop plants such as oats, wheat, cotton has been considerable been increased by sustainable plant breeding efforts in selecting and developing varieties with improved transport of assimilates to edible or economically important portions of the plant.
- **Factors controlling translocation and assimilates partitioning in higher plants**
    - Competition among sink tissues for available translocated assimilates
    - Photosynthesis and Sink demand
    - Long distance signals between sources and Sinks
    - Plasmodesmata.

6.4 Relationship between the source and sink

- Crop species differ in yields but when environmental conditions are conductive, their yield limitations can be caused by limitations of the;
    - Source
    - Sink
    - Source and sink
- We can detect that the limitation in crop yields is caused by sinks using;
    - Shading experiments
    - Defoliation experiment
    - Stay green phenomenon of leaves
    - Solution sink limitation can be solved by plant breeders who can increase the sink capacity.
- We can detect that the limitation in crop yields is caused by the source using;

- Observation;
  - Leaves senesce (dry and die) before the black layer formation in case of grain crops.
  - Plant lodging during reproduction and maturation phase.

- We can detect that the limitation in crop yields is caused by both source and /or sinks
  - This may be difficult to establish because feedback mechanism to stress conceals the effect of the limiting factor.
    - To avoid stress, experiments are carried under optimum conditions
  - How to know;
    - F: G ratio is used to determine if it is the source or sink is limiting. Where F = total CHO during flowering period and G= total CHO during filling period
    - The F determines the potential size of a sink formed
    - The G determines the ultimate size of sinks developed
    - Once ratio is interpreted, a solution is instituted.

TOPIC 7: PLANT NUTRITION

7.1 Essential elements

Below is the number of essential elements required to support life;

- Human beings (19): C, H, O, N, P, K, Ca, Mg, S, Fe, Mn, Zn, Cu, Mo, Cl, Na, I, Co, and F.
- Farm animals (18): C, H, O, N, P, K, Ca, Mg, S, Fe, Mn, Na, Zn, Cu, Mo, Cl, I, and Co.
- Plants (16): C, H, O, N, P, K, Ca, Mg, S, Fe, Mn, Zn, Cu, Mo, Cl and B.

7.2 Criteria for a mineral nutrient to be considered an essential element

- A plant nutrient is essential when the nutrient:
    - Element is needed for the plant to complete growth cycle.
    - Element performs a function that cannot be replaced by another element.
    - Element is directly or indirectly involved in plant metabolic processes.
- Proper nutrition is essential for satisfactory crop growth and reproduction.
- Air and water supply three essential elements; carbon, hydrogen and oxygen. Thus these three are commonly referred to as non- mineral elements.
- The soil or fertiliser supply macro (required in large quantities) and micro (required in small quantities);
    - Six (6) elements are needed in large quantities (Nitrogen (N), Phosphorous (P), Potassium (K), Sulphur (S), Calcium (Ca) and Magnesium (Mg)).
    - Eight (8) are needed in small quantities (Iron (Fe), Manganese (Mn), Copper (Cu), Zinc (Zn), Boron (B), Molybdenum (Mo), Chlorine (Cl) and Nickel (Ni)).
    - With the exception of hydrogen, carbon, and oxygen all other inorganic plant requirements are obtained directly or indirectly from the soil minerals, hence these elements are called mineral nutrients.
    - Strictly speaking, nitrogen is not a mineral element but it has been included in the list because it can also be obtained by plants from soil.
- The mineral elements are taken by plants from soils mostly in the form of ions as either a cation, a positively-charged ion, or an anion, a negatively-charged ion. The ions are taken up by plants through:
    - Roots from the soil solutions, exchangeable ions on the surface of clay and humus particles and readily decomposable minerals.
    - The leaves.

7.3 Classification of plant nutrients

7.3.1 Based on amount required by the plant (supply to maximise crop yields)

Depending on the quantity of nutrients present in plants, they can be grouped into three:

- **Basic Nutrients**
  - The basic nutrients are carbon, hydrogen and oxygen, constitute 96% of total dry matter of plants.
  - Carbon and oxygen constitute 45%.
  - The total dry matter produced by rice crop in one season is about 12 ton/ha. In this 5.4 t is carbon, 5.4 ton is oxygen and 0.7 ton is hydrogen.
- **Macronutrienls**
  - The nutrients required in large quantities are known as macronutrients.
  - They are N, P, K, Ca, Mg, and S. Among these, N, P and K are called primary nutrients and Ca, Mg and S are known as secondary nutrients.
  - The later are known as secondary nutrients as they are inadvertently applied to the soils when N, P and K fertilizers, which contain these nutrients, are used.
- **Micronutrienls**
  - The nutrients which are required in small quantities are known as micronutrients or trace elements.
  - They are Fe, Zn, Cu, B, Mo and Cl.
  - These elements are very efficient and minute quantities produce optimum effects.
  - On the other hand, even a slight deficiency or excess is harmful to the plants.

7.3.2 Based on role of element in plant systems

- Structural elements include carbon (C), Hydrogen (H) and Oxygen (O).
- Accessory structural elements, includes nitrogen (N), phosphorous (P) and sulphur (S).
- Regulator and carriers, include potassium (K), calcium (Ca) and magnesium (Mg).
- Catalysts and activators, include iron (Fe), manganese (Mn), zinc (Zn), copper (Cu), molybdenum (Mo), chlorine (Cl) and boron (B)

7.3.3 Based on relative abundance in the plant tissues

This classification recognises that macro nutrients are absorbed by plants in more proportions than in micro-nutrients. Macro- nutrients are reported in % while micro-nutrients in mg per kg DM. This classification also recognises three groups with the last two groups having subgroups:

- Non- mineral elements: C, H and O.
- Macro-nutrients:
    - Primary: N, P and K
    - Secondary: Ca, Mg and S
- Micro-nutrients:
    - Required by all plants: Fe, Mn, Zn, Cu, B, Cl and Mo
    - Required by some plants: Na, Ni, Co, Si and V

AVERAGE CONCENTRATION RANGES OF ESSENTIAL NUTRIENT ELEMENTS IN CROPS

Macro-elements	Concentration range percentages in DM
O	45
C	45
H	6
N	0.1 – 6.0
P	0.01 – 0.7
K	0.2 – 6.0
Ca	0.2 – 1.0
Mg	0.1 – 0.4
S	0.1 – 0.4
Si	0.2 - 2.0
Cl	0.2 – 2.0
Na	0.01 – 10

Microlements	Concentration range mg per kg in DM
Fe	50 – 250
Mn	20 – 500
Zn	25 – 150
Cu	2 – 20
B	6 – 60
Mo	± 1
Co	0.02 – 0.5
Ni	0.05
V	± 1

7.3.4 Based on chemical nature according to periodic table

The nutrients can be classified into cations and anions and metals and non-metals based on their chemical nature.

- Cations: K, Ca, Mg, Fe, Mn, Zn, Cu,
- Anions: NO_3, PO_4, SO_4
- Metals: K, Ca, Mg, Fe, Mo, Zn, Cu,
- Non-metals: N, P, S, B, Mo, Cl

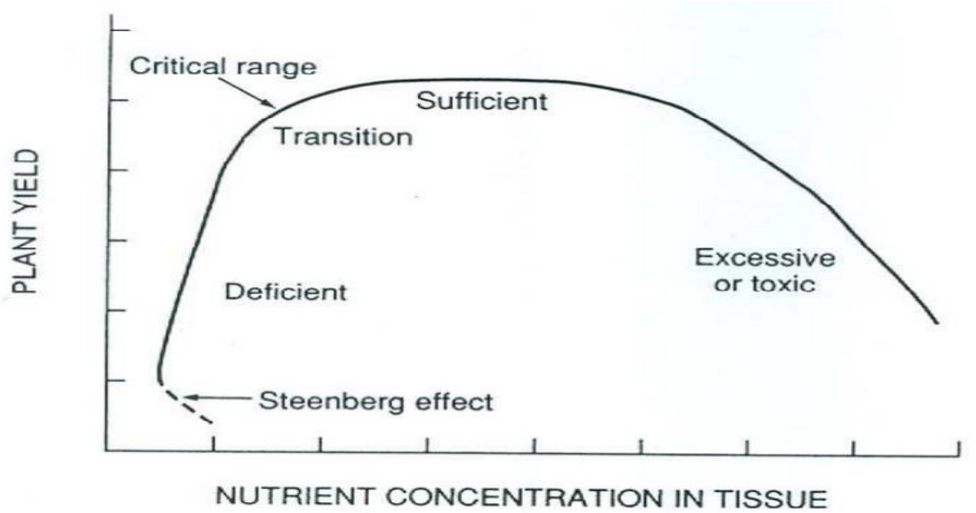
7.3.5 Based on mobility of nutrients

This classification recognises that essential elements can be mobile in the soil or in the plant.

- **Mobility of nutrients in the soil**
    - Has considerable influence on availability of nutrients to plants and method of fertilizer application. For plants to take up these nutrients, two processes are important:
        - Movement of nutrient ions to the absorbing root surface, and
        - Roots reaching the area where nutrients are available.
    - Based on the mobility in the soil, the nutrient ions can be grouped as mobile, less mobile and immobile.
    - In the case of immobile nutrients, the roots have to reach the area of nutrient availability and forage volume is limited to root surface area.
    - For highly mobile nutrients, the entire volume of the root zone is forage area.
- **Based on mobility in Plants**
    - Knowledge of the mobility of nutrients in the plant helps in finding what nutrient is deficient.
    - A mobile nutrient in the plant moves to the growing points in case of deficiency.
    - Deficiency symptoms, therefore, appear on the lower leaves.
    - Four groups are known
        - highly mobile
        - Moderate mobile
        - Less mobile
        - Immobile

7.4 Relationship between nutrient concentration in the plant tissue and pant yield

- The relationship is better understood using a graph like shown below**.**



7.5 Mechanism of nutrient transport to plant root surface

- Nutrients must reach the surface of a root for plant uptake of essential elements to occur. There are three major mechanisms of movement of ions from soil to roots:
    - Root interception
    - Mass flow
    - Diffusion
- Nutrient uptake rates of three mechanisms are related to the chemical characteristics and behaviour of the nutrient element in soil. Using maize, Havlin *et al*. (2005) reported variable percentages in the supply of the major plant nutrients as tabulated below:

Nutrient	**Nutrient uptake mechanism**		
	Root interception	**Mass flow**	**Diffusion**
	Supply (%)		
Nitrogen	1	99	0
Phosphorus	2	4	94
Potassium	2	20	78
Calcium	12	88	0
Magnesium	27	73	0
Sulphur	4	94	2

7.6 Transportation of mineral nutrients from root surface to xylem vessels

- In **symplastic pathway**, mineral nutrients entering the cytoplasm of the epidermis move across the cytoplasm of the cortex, endodermis of pericycle through plasmodesmata and finally reach the xylem. In short solute movement is **from cell to cell** through the **plasmodesmata**, which **by-passes the vacuoles**.
- **In Apoplastic pathways**, the mineral ions, which enter the cell wall of the epidermis move across cell wall of cortex, cytoplasm of endodermis, cell walls of pericycle and finally reach the xylem. In short, the solute (mineral nutrients) movement is through the **extracellular space** (cell wall and intercellular spaces).

7.7 Translocation mineral nutrients

- Most soil nutrients are translocated through the xylem to the growing shoots points.

- Translocation is assisted by transpiration and water movement.

7.8 Macro nutrients

- Due to the amounts required, macronutrients can either be supplied in excess, adequate or low quantities.
- Macronutrients take two forms; mineral and non-mineral nutrients.

7.8.1 Non-mineral nutrients

- Only three essential elements (C, H, and O) belong to this group.
- Carbon is absorbed in the form of CO_2 from the air during photosynthesis and is used to synthesise carbohydrates.
- Hydrogen and oxygen are absorbed in from of water and used together with carbon to synthesise all organic components.
- Carbohydrates are in turn converted to other organic compounds.
- **Functions**
    - Forms the backbone of organic molecules or components in plants.
    - In all organic molecules; carbohydrates, proteins (including enzymes), lipids, nucleic acids, vitamins and hormones, etc. carbon, oxygen and hydrogen are essential elements.
- **Deficiency**
    - Absence of the three elements results in ultimate death of the plant.

7.8.2 Nitrogen

- Absorbed as NO_3^- (in general) and NH_4^+.
- **Functions**
    - Is a component of many essential components (proteins and enzymes).
    - Plays an extremely important role in the growth of plants (especially the above ground biomass).
    - Contributes to the green colour in plants due to its stimulating effect on chlorophyll production, the primary absorber of light energy needed for photosynthesis.
    - Governs the utilisation of P, K and other elements.
    - Supports the growth of soil microorganisms.
- **Deficiency**
    - Nitrogen shortages in soils are more general than for any other essential elements due to its high absorption by plants and the fact that it is easily leached from soil by water.

- Mild restricts plant growth, but often in a subtle manner that can only be assessed by comparison to plants grown with an adequate supply.
- Moderate cause leaves to be light green or yellowish.
- Severe causes necrosis (tissue death) starting at the tip of old leaves, with the tissue death developing a V-Pattern down the midrib toward the base of the leaf.

- **Toxic level**
  - Occurrences of yield losses.
  - Stimulates excessive vegetative growth, dark green leaves, lodging, maturity is delayed with increases susceptibility to pest and disease.
  - Lengthening of crop duration and narrow leaves.

7.8.3 Phosphorous

- Besides nitrogen, phosphorous is the second most limited factor in soils.
- Is absorbed primarily in the monovalent form ($H_2PO_4^-$) but slower in the divalent form (HPO_4^{2-}).
- Soil pH determines absorption; $H_2PO_4^-$ is greater at low pH while HPO_4^{2-} is greater at higher values of soil pH.
- Thus, the availability of P (< 7 favours the monovalent and >7 the divalent form).
- **Note:** P is never reduced in plants but remains the same form (as Pi or bound to organic molecules).
- **Functions**
  - It is essential for cell division and development.
  - It stimulates early root development and growth and there by helps to establish seedlings quickly.
  - It is an essential component of sugar phosphates
  - It is a component of nucleotide units (DNA and RNA).
  - Is a component of phospholipids in membranes.
  - Plays an essential role in energy metabolism due to its presence in the composition of ATP, ADP, AMP, Pi (free phosphate) and PPi (pirophosphate).
  - It gives rapid and vigorous start to plants strengthen's straw and decreases lodging tendency.
  - It is essential for seed formation because larger quantities of P is found in seed and fruit-phytic acid is the principle storage from of phosphorus in seeds.
  - It increases the activity of **rhizobia** and increases the formation of root nodules
- **Deficiency:**
  - Phosphorous is mobile in plants and when a deficiency occurs it is translocated from older tissues to the active meristematic regions. However, since most soils are deficient;

- It arrests metabolism resulting in reduction of total nitrogen of plants.
- Reduced sugar content and cause stunted growth and late maturity.
- Cause premature leaf fall.
- Develops necrotic area on the leaf petiole and in the fruit.
- Leaves will show characteristic bluish green colour.
- Severity causes reddening of leaves in grasses like maize.

- **Toxic level**
  - Profuse root growth i.e. laterals and fibrous root lets.
  - It develops normal growth having green leaf colour.
  - It may cause in some cases trace elements deficiencies i.e. Zinc and Iron.

7.8.4 Potassium

- Can also become limited in soil, although soils tend to contain sufficient potassium.
- Absorbed as K^+
- **Function**:
  - It is an inorganic co-factor that activates many enzymes involved in photosynthesis and respiration.
  - Also activates enzymes involved in starch and protein synthesis.
  - Plays an important role in stomata movement i.e regulates opening and closing.
  - Keeps cell's osmotic potential and therefore turgor state intact.
  - There are strong indications that potassium plays a role in the translocation of organic compounds (e.g. Sucrose) over membranes but clarity on this is still awaited via future research.
  - Essential for development of chlorophyll.
  - It improves vigour of the plants to enable to with stand adverse climatic conditions.
  - Reduces lodging in cereal crops.
  - It regulates the movement of ions with in the plants and hence it is called traffic policeman of the plant.
- **Deficiency**
  - No glaring visual symptom due to mobility in phloem.
  - However, when it is limited the following could be seen:
  - Plant becomes stunted in growth with shortening of internodes.
  - Reduced rate of photosynthesis.
  - Chlorosis, yellowing of leaves and leaf scorch in case of fruits trees.
- **Toxic level**

- Usually not absorbed in excessive by plants.
- Excess potassium can aggravate the uptake of magnesium, manganese, zinc and iron.

7.8.5 Sulphur

- Absorbed by plants as SO_4^{2-}.
- **Functions**
  - Is a component of many proteins and is an essential component of the two amino acids Cystein and Methionine.
  - Vitamins such as Thiamin and Biotin as well as Co-enzyme A [essential for respiration and the breakdown of lipids (fats)] contains sulphur in their composition.
- **Deficiency**
  - No glaring visual symptom due to mobility in phloem.
  - Leaf necrosis from the tip and the leaf margin.
  - Yellowing of whole plant.
  - Stunted growth.
- **Toxic level**
  - Leaf size will be reduced and overall growth will be stunted.
  - Leaves yellowing or scorched at edges.

7.8.6 Calcium

- Most soils contain sufficient calcium to support normal plant growth.
- Absorbed in the divalent form Ca^{2+}.
- **Functions**
  - Is essential for normal membrane functions in all cells by binding phospholipids together and also to membrane proteins.
  - Plays a role in cell wall formation and increases in stiffness of plants.
  - It is immobile in plants and hence the deficiency is observed in younger leaves.
  - Promotes root development and growth of plants, root elongation and cell division.
  - Helps to translocate the sugar in the plants.
  - It involves chromosome stability and that it is a constituent of chromosome structure.
  - Encourages seed production.
  - Activates enzyme phosphate and kinease.

- It binds DNA to protein molecules.

- **Deficiency**
  - No glaring visual symptom due to availability in soils.
  - However, when supply is limited the following will be observed:
  - Severity in groundnuts causes pops.
  - Young leaves of terminal buds dieback at the tip and margins or remain folded in cereals.
  - Normal growth in affected.
  - Root may become short, stubby and brown.
  - Causes acidity of soil.
  - Cell may become rigid and brittle.
  - Young leaves of cereals remain folded.

- **Toxic level**
  - Difficult to distinguish visually.
  - May precipitate with sulfur in solution and cause clouding or residue in tank.

7.8.7 Magnesium

- Relatively abundant in soil and is seldom a limiting factor.
- Absorbed by plants in the divalent form Mg^{2+}.
- **Functions**
  - Forms part of the chlorophyll molecule and is therefore essential for chlorophyll production.
  - Imparts dark green colour in leaves.
  - Serves as a structural component in ribosomes and stabilizing the ribosome configuration for protein synthesis.
  - Involves numbers of physiological and biochemical function.
  - Activates phosphorylating enzymes in CHO metabolism.
  - Act as a cofactor for certain enzymes other than PO_4 transfer enzyme.
  - Increases the oil content of oil seed crops.
  - Regulates the uptake of other nutrients.

- **Deficiency**
  - Relatively rare as far as plants are concerned. However, in extreme situations;
  - Leaf necrosis from the tip and the leaf margin or yellowing between the leaf veins and around the leaf margins in lower and older leaves.

- Leaf margins may curve upward and may turn red-brown to purple in some plants.
- Interveinal chlorosis of the leaf in which only the leave veins remain green.
- Stiff brittle, twisted leaves, wrinkled and distortion of leaves.

- **Toxic level**
  - Magnesium toxicity is rare and not generally exhibited visibly.

7.9 Micro nutrients

7.9.1 Iron

- Absorbed by plants in the divalent (Fe^{2+}) and trivalent Fe^{3+} form and as chelated ions.
- Iron is not very mobile in soil as well as in old leaves.
- **Functions**
  - It helps in the synthesis of chlorophyll – specific enzymes involved are activated by Fe^{2+}.
  - Structural component of porphyrin molecules like cytochromes, hematin, hemes, ferrichrome and hemoglobin.
  - These substances are involved in oxidation-reduction reactions in respiration and photosynthesis. Thus, acts as electron transfer agent and changes from the Fe^{2+} to the Fe^{3+} form and vice versa.
  - Structural component of non-hemine compounds like ferredoxins.
  - Constituent of enzyme systems e.g. cytochrome oxidase, catalase, nitrogenase reaction in plants.
  - Component of flavoprotein like FMN = Flavin mono nucleotides and FAD = Flavin Adinosine Dinudeotide.
- **Deficiency**
  - Iron chlorosis, which causes leaves to turn yellow between the veins.
  - Mild only seen in new formed leaves since Fe is immobile element within plant.
  - Moderate, the whole plant may turn yellow or white and may produce fewer shoots than normal.
  - It occurs in soils of calcareous or alkaline soils and poorly drained water logged soils.
  - Severity cause susceptibility to sunburn, entire leaf turning yellow which leads dead areas at shoot tips and to leaf drop.

7.9.2 Chlorine

- Absorbed by plants as the chloride ion Cl^- through roots and aerial parts.
- Is absorbed 10 to 100 times more than what the plant needs. The reason for this still needs clarification.
- **Note**: Is a component of 4-chloro-indole acetic acid, a natural auxin hormone.

- **Functions**
    - Stimulates the splitting of water (hydrolysis) during the light reaction of photosynthesis.
    - Essential for biochemical reactions
    - It is an important osmotic active solute that improves water absorption by cells.
    - Act as a counter ion during rapid K fluxes.
    - Involves in the evaluation of O_2 in photosynthesis.
    - Essential for cell division in leaves.
- **Deficiency**
    - Chloride deficiencies are not common and are typically limited to areas with high rainfall and sandy soils.
    - Identifying chloride deficiency in plants is difficult because the symptoms of deficiency are similar to the symptoms of many cereal plant diseases (reduction in growth).
    - Seen as chlorosis of the younger leaves and wilting of the plant.
    - It causes crops like cabbage lack its distinctive smell.

7.9.3 Molybdenum

- Found in soil in the form of molybdate (MoO_4^{2-}) but also as MoS_2 (molybdenum sulfide).
- In the first case molybdenum is in the Mo^{6+} ion form and in the last case in the Mo^{4+} form.
- Very little is known about the form in which plants absorb molybdenum, perhaps because it is needed in such small quantities.
- **Functions**
    - Most important physiological function is that it forms part of the enzyme nitrate reductase and therefore plays a role in the reduction of nitrate (NO^{3-}) to nitrite (NO^{2-}).
    - Structural component of nitrogenase enzyme - involved in N fixation.
    - Essential for absorption and translocation of Fe in plants.
    - Also plays a role in the conversion of abscissic acid aldehide to the hormone ABA (abscissic acid). Because of this the possibility exists that it plays a role in drought resistance in plants.
    - More research is needed.
- **Deficiency**
    - Since the most important function of molybdenum in plant metabolism is in the NO_3^- (nitrate) reduction, molybdenum deficiency resembles nitrogen deficiency (general chlorosis and stunted growth).
    - Inhibits flower formation

- Imbalances various amino acids in plants.
- Reduce activity of symbiotic and non- symbiotic N fixation.

7.9.4 Nickel

- Absorbed by plants in the form of Ni^{2+}.
- Recent research has shown that nickel should probably be added to the essential micro-element list.
- **Functions**
  - Forms part of the enzyme urease that breaks down urea to CO_2 and $NH4^{+}$.
  - It seems further that urea production takes place in all plants as part of their normal physiology.
  - Without nickel toxic amounts of urea can build up in leaves, especially old leaves when the plant starts to flower.
  - It has also been shown that nickel can stimulate growth in wheat, tomatoes and barley.
  - Note: The above “facts” support the “essential-status” of nickel.
- **Deficiency**
  - Necrotic lesions or dead areas on the leaf.
  - Inhibition of seed germination in future generations.
  - Research is needed to test the effect on plants.
- **Toxic levels**
  - High levels of Ni may induce Zn or Fe deficiency.

7.9.5 Copper

  - Absorbed by plants as cupric ion (Cu^{2+}) in well aerated soil or the monovalent form (Cu^{+}) in wet soils containing little oxygen.
  - Normal concentration in plants is 5 - 20 ppm.
- **Functions**
  - Important for many enzyme controlled oxidation and reduction reactions where it acts as inorganic enzyme co-factor. It acts as an electron carrier.
  - Best example is the enzyme Cytochrome oxidase, a respiratory enzyme in the mitochondrion, and plastosianin, a chloroplast protein.
  - Essential for the synthesis of vitamin A
  - Act as a catalyst in respiration
  - Constituent of chlorophyll.

- **Deficiency**
    - Plants seldom experience a copper deficiency because they need so little.
    - However, when deficiency occurs;
    - Chlorosis, withering and distortion of terminal buds.
    - Dead tissue appears along the tips and edges of leaves.
    - Multiple bud formation in the leaf axil and malformation of leaves.
    - "Die-back" disease in young leaves of citrus and cracking in guava fruits.

7.9.6 Zinc

- Absorbed as Zn^{2+}.
- Normal concentration in plant is 25 to 150 ppm, deficiency of < 20 ppm and toxic level of > 400 ppm.
- **Functions**
    - Important for chlorophyll production as well as the prevention of chlorophyll breakdown.
    - Involved in the synthesis of the growth hormone auxin (IAA) and is therefore also seen as essential for normal growth.
    - Acts as co-factor that activates more than 80 enzymes.
    - Helps in reproduction of certain plants.
    - Stabilise rhibosomal fractions
- **Deficiency**
    - Small and distorted leaves.
    - Chlorosis between veins which starts from the bottom of the leaf.
    - Short internoded and stunted growth.
    - Roset small fruit in apples and peaches (i.e malformation of fruits).
    - Death of tissue, discolored
    - Reduced growth hormone production.

7.9.7 Manganese

- Absorbed as Mn^{2+} but is found in 3 oxidized forms; Mn^{2+}, Mn^{3+} and Mn^{4+}.
- Can also remain in soil in an insoluble form.
- **Functions**
    - Needed for normal development of thylakoids (grana membranes) in the chloroplast and possibly the whole membrane system of the chloroplast.

- Involved in oxidation - reduction - process in decarboxylation and hydrolysis reactions.
- Involved in enzyme systems and various enzyme reactions in the citric acid cycle.
- It is a substitute for Mg^{2+} in many of the phosphorylating and group transfer reactions.
- Acts as metal activator for many enzymes.

- **Deficiency**
  - Deficiencies are remarkably similar to magnesium because both are involved in photosynthesis.
  - Leaves become yellow and there is also interveinal chlorosis.
  - Manganese is immobile in a plant so chlorosis starts with young leaves.
  - Deficiency increases asparatic acid and decreases glutamine
  - Increases respiration
  - Accumulation N compounds mainly as amines.

7.9.8 Boron

- Absorbed from soil as boracic acid (H_3BO_3) in an undissosiated form.
- **Functions**
  - Essential for DNA and RNA synthesis.
  - Plays a role in cell division in root and stem growth tips.
  - Necessary for pollen vitality, i.e. to germinate and to extend the pollen tube. New cell development in meristematic tissue. Thus, facilitates Pollination, fruit / seed set.
  - Translocation of sugars, starches, N and P.
  - Synthesis of proteins and amino acids.
  - Nodule formation in legumes.
  - Regulation of carbohydrates metabolism.
  - More research is needed to elucidate the biochemical involvement of boron in plants.
- **Deficiency**
  - Inability to extend root tips in other plants.
  - Since it is immobile, deficiency symptoms occurs in terminal bud growth.
  - Flowering and fruit development are restricted.
  - Sterility and mal formation of reproductive organs.
  - Thickened and curled leaves.
  - Discoloration, cracking or rotting of fruit, tubers or roots.

7.9.9 Sodium

- Absorbed as monosilicic acid (Na^{+}).
- **Functions**
    - Essential for maintaining turgor and growth of Plants.
    - Helps oxalic acid accumulation in Plants.
    - Helps in stomatal opening and regulate NO_3^- reductase activity**.**

7.9.10 Silica

- Absorbed as $Si(OH)_4^o$
- **Functions**
    - Contributes structure of cell wall.
    - Contributes drought tolerance to crops.
    - Regulates photosynthesis.
    - Deactivates invertase enzyme activity in sugarcane resulting greater sucrose production.
    - Larger amounts of Si are accumulated in intracellular deposits known as 'PLANT OPALS'.
    - Increased available P.
    - Responsible for plant disease resistance.

7.9.11 Vanadium

- Absorbed as V^{+}
- **Functions**
    - Low concentration of Vanadium is beneficial for growth of microorganisms.
    - Vanadium substitute for Mo in N fixation by Rhizobia.
    - Involves in biological - oxidation-reduction reactions.
- **Deficiency**
    - Restricts growth of microorganism
    - Reduction in biological - oxidation-reduction reactions

7.9.12 Cobolt

- Absobed as Co^{2+}.
- **Functions**
    - Growth of symbiotic microorganisms such as symbiotic rhizobium, free-living N_2-fixing bacteria, and blue-green algae.

- It forms vitamin B12 during growth and development of symbiotic microoganism.
- Improves growth, transpiration and photosynthesis.
- Take parts in leghemoglobin metabolism and Rhibonucliotide reductase.

- **Deficiency**
  - Affects nitrogen fixation in soil.

TOPIC 8: PLANT HORMONES AND GROWTH

8.1 Definition of hormone

Hormones are naturally occurring, organic molecules that, at low concentration, exert a profound influence on physiological processes. In addition, hormones, as defined by animal physiologists, are;

- Synthesized in a discrete organ or tissue, and
- Transported in the bloodstream to a specific target tissue where they
- Control a physiological response in a concentration-dependent manner.

Plant hormones usually called phytohormones are naturally occurring organic substances that profoundly influence physiological processes at low concentration. The site of synthesis and mode of transport for plant hormones, however, is not always so clearly localized. Although some tissues or parts of tissues may be characterized by higher hormone levels than others, synthesis of plant hormones appears to be much more diffuse and cannot always be localized to discrete organs.

What are phytohormones? Plant scientists explain phytohormones as;

- Chemicals produced in one tissue or organ.
- Transported from synthesising organ to organ of action.
    - Chemicals diffuse from cell to cell.
    - Maybe carried by water if aquatic to other cells in same individual.
    - Higher plants transport the chemical in fluid using the vascular systems.
- Has its effect in another tissue or organ (target cell or organ).
- Coordinate and control or regulate all plant actives (growth, reproduction, development and death).
    - Make sure activities throughout the plant are timed properly.

8.2 Auxins

Auxins are a group of hormones that regulate growth in many parts of a plant. The first plant growth hormone to be discovered by Charles Darwin and son in 1880. The principal natural auxin is indole-3-acetic acid. Natural auxin biosynthesis is associated with rapidly dividing and growing tissues, especially in shoots.

Biosynthesis

- It is synthesized in meristems, young leaves, and developing fruits and seeds.

- Although virtually all plant tissues appear to be capable of producing low levels of auxin, shoot apical meristems and young leaves are the primary sites of auxin synthesis.
- Root apical meristems are also important sites of auxin synthesis, especially as the roots elongate and mature, although the root remains dependent on the shoot for much of its auxin. Young fruits and seeds contain high levels of auxin, but it is unclear whether this auxin is newly synthesized or transported from maternal tissues during development.
- However, shoot apexes synthesize more auxin than the root apecies.

Transportation

- The main axes of shoots and roots, along with their branches, exhibit apex-base structural polarity, and this structural polarity is dependent on the polarity of auxin transport.
- IAA moves mainly from the apical to the basal end (basipetally).
- This type of unidirectional transport is termed polar transport.
- Auxin is the only plant growth hormone that has been clearly shown to be transported polarly, and polar transport of this hormone is found in almost all plants.

Effects of auxin on growth and development

- Promotes phototropism
- Promotes geotropism
- Promotes apical dominance
- Stimulates fruit development
- Leaf abscission and fruit drop
- Promotes adventitious root development

8.3 Gibberellins

This is the phytohormone best known for its promotion of stem elongation.

Biosynthesis

- Gibberellin biosynthesis occurs in multiple plant organs, like germinating embryos, young seedlings, shoot apices, developing'- seeds, -leaves, and -roots and even in some fungi.
- It is generally accepted that there are three principal sites of gibberellin biosynthesis:

- Developing seeds and fruits,
- The young leaves of developing apical buds and elongating shoots, and
- The apical regions of roots.
- However, immature seeds and fruits are also prominent sites of gibberellin biosynthesis.

Transportation

- The gibberellins are mobile and may act either locally or distant from their sites of synthesis.
- Gibberellins are transported in both phloem and xylem saps.

Effects of gibberellins on growth and development

- promotes stem elongation
- gibberellins may also be involved in plant movements and flowering
- Gibberellins promote seed germination via interrupting of dormancy
- They promote early seed development and germination
- They regulate the transition from juvenile to adult phase
- They have influence on floral initiation and sex determination
- GAs promote pollen development and tube growth
- Gibberellins promote fruit set and parthenocarpy

8.4 Cytokinins

Naturally occurring cytokinins are all **adenine derivatives** with either an isoprene-related side chain or an aromatic (cyclic) side chain. The former are called isoprenoid cytokinins and the latter are called aromatic cytokinins. Although there is some variation depending on species and developmental stage, the most common isoprenoid cytokinins are N6-(2-isopentenyl)-adenine (iP), trans-zeatin (tZ), and dihydrozeatin (DZ). The aromatic cytokinins, such as benzyladenine (BA) are less common and are found in only a few species.

Biosynthesis

- Cytokinins are synthesized in roots, developing embryos, young leaves, fruits.
- A major site of cytokinins biosynthesis in higher plants is the root.
- High cytokinins levels have been found in roots, especially the mitotically active root tip, and in the xylem sap of roots from a variety of sources.

- It is generally concluded that roots are a principal source of cytokinins in most, if not all, plants and that they are transported to the aerial portion of the plant through the xylem. The delayed senescence when roots are allowed to form is apparently due to the presence of cytokinins, which are synthesized in the root and transported to the leaf through the vascular tissue.

Effects of cytokinins on growth and development

- Promote shoot growth by increasing cell proliferation (cell division and differentiation) in the shoot apical meristem.
- Inhibit root growth by promoting the exit of cells from the root apical meristem.
- Both cytokinins and auxin regulate the plant cell cycle and are needed for cell division.
- The auxin: cytokinin ratio regulates morphogenesis in cultured tissues.
- Cytokinins modify apical dominance and promote lateral bud growth.
- Cytokinins delay leaf senescence, promote nutrient mobilization, help regulate the synthesis of photosynthetic pigments and proteins.
- Cytokinin-overproducing plants have delayed senescence and yield more grain.

8.5 Ethylene

Ethylene is the only gas hormone with a single representative. Ethylene is apparently not required for normal vegetative growth, although it can have a significant impact on the development of roots and shoots. Ethylene appears to be synthesized primarily in response to stress and may be produced in large amounts by tissues undergoing senescence or ripening. It is commonly used to enhance ripening in bananas and other fruits that are picked green for shipment as well.

Bbiosynthesis

- Can be produced by almost all parts of higher plants and occurs in all plant organs – roots, stems, leaves, bulbs, tubers, fruits, seeds, and so on – although the rate of production may vary depending on the stage of development.
- Ethylene production will also vary from tissue to tissue within the organ, but is frequently located in peripheral tissues.

Effects of ethylene on plant growth and development

- Ethylene affects the transcription of numerous genes via specific transcription factors
- The hormone promotes the ripening of some fruits
- Ethylene inhibits hypocotyl elongation
- It regulates flowering and sex determination, in some species.
- Regulates defence responses
- Ethylene is active in leaf and flower senescence and in leaf abscission.

8.6 Abscisic acid

The primary functions of abscisic acid are

- Prohibiting precocious germination and promoting dormancy in seeds
- Inducing stomatal closure and the production of molecules that protect cells against desiccation in times of water stress.

Biosynthesis

- Shoot apex (terminal bud) produce abscisic acid in higher plants especially the leaves.

Transportation

- Phloem transport to the roots and cycled back via xylem to the leaf to close the stomata.

Effects of abscisic acid on growth and development

- Promotes dormancy in woody stems as winter approaches, woody plants cease growth and develop protective bud scales.
- Promotes seed dormancy, seeds have a high concentration of abscisic acid and cannot germinate until these levels are reduced
- Acts as plant stress hormone and protects cells from drought, freezing, salination.
- In seed development, abscisic acid promotes the synthesis of storage proteins and lipids, as well as special proteins
- abscisic acid greatly accelerates the senescence of leaves, thereby increasing ethylene formation and stimulating abscission

8.7 Brassinosteroids

Brassinosteroids are steroid hormones with a chemical structure similar to the steroid hormones in animals. Brassinosteroids elicit an impressive array of developmental responses, including an increased rate of stem and pollen tube elongation, increased rates of cell division (in the presence of auxin and cytokinin), seed germination, leaf morphogenesis, apical dominance, inhibition of root elongation, vascular differentiation, accelerated senescence, and cell death. Brassinosteroids are also implicated in mediating responses to both abiotic and biotic stress, including salt, drought, temperature extremes, and pathogens.

Biosynthesis

- Roots, shoots and leaves synthesis.

Transportation

When synthesised in roots can be transported via xylem to the shoots but are poorly translocated out of leave.

Effects of brassinosteroids **on growth and developmental**

- Promote both cell proliferation and cell elongation
- Promote root growth at low concentrations and inhibit root growth at high concentration
- Promote differentiation of the xylem and supress that of the phloem
- Promote seed germination by interacting with their hormones, such as gibberellins and abscisic acid

TOPIC 9: SOME SPECIFIC COMMON CROPS GROWN IN ZAMBIA

- Students to refer to corrections done during presentation of the common crops grown in Zambia as they read the assignments
- The questions was :

A field crop has been allocated to each group and you are required to;

a. Write the scientific name of the crop.
b. Mention the crop's origin.
c. Name any two Zambian varieties suitable for each agro- ecological region and their potential yield.
d. Write short notes on the following agronomic practices if higher yields are to be obtained by a farmer.
    - Land preparation
    - Soil and fertiliser requirements.
    - Temperature requirements
    - Water requirement
    - Planting and plant population.
    - Weed management
    - Four major insects and diseases each and how they can be controlled
    - Harvesting
    - Post-harvest management
    - Marketing

9.1 Maize

9.2 Cassava

9.3 Sunflower

9.4 Cotton

9.5. Common beans

TOPIC 10: PLANT POPULATIONS

- Students to refer to corrections done during presentation of the of the assignment as they read the assignments

- The questions were;
  I) Outline the growth parameters that are used to determine the inter-row and intra-row spacing in crops.
  II) You have a field which is 700 m in length and 500 m in width, supposing the intra-row spacing is 35 cm and the inter-row spacing is 50 cm. calculate
    - Number of rows the field will have.
    - Number of planting stations each row will have.
    - What is the plant population?
  III) Define
    - Intra-row spacing
    - Inter-row spacing
    - Plant population

TOPIC 11: FIXATION OF NITROGEN BY LEGUMINOUS CROPS

- Students to refer to corrections done during presentation of the assignment as they read the assignments

- The questions were
    - Outline the steps in the formation of root nodules in leguminous plants.
    - Outline the steps in the fixation of nitrogen by symbiotic microorganisms in legumes.

TOPIC 12: PROTECTED AGRICULTURE AND ORGANIC AGRICULTURE

- Students to refer to corrections done during presentation of the assignment as they read the assignments.
- The questions were;

12.1 Protected agriculture

Crop Scientists are determined in producing food to feed the ever increasing human population in the world. Now any crop can be grown and produced throughout the year in any environment through protected agriculture technology.

a. What is protected agriculture (PA)?
b. Outline the various types of PA structures and list any food crops which can be produced in them in Zambia.
c. Outline the benefits of PA.
d. Outline some reasons why the majority of Zambian farmers are not using PA technology.
e. List the structural materials used to make PA structures and comment on their availability in Zambia.

12.2 Organic agriculture

The international Federation of organic agriculture movement (IFOAM) has been lobbying farmers in the world to stop using any form of synthetic chemical in food production.

a. Outline any four ways farmers could benefit if they accepted to convert the food production from conventional to organic farming system
b. Outline any four ways options promoted by IFOAM to overcome yield reduction as farmers stop using synthetic chemicals in agriculture.
c. Outline any four reasons why Zambian farmers are not converting to organic crop production.

TOPIC 13: METABOLISM

- Students to refer to corrections done during presentation of the of the assignment as they read the assignments
- The question was;

a. Outline the external and internal factors that control photosynthesis.
b. Outline the steps of light reaction during photosynthesis.
c. Outline the phases of dark reaction (Calvin cycle) in photosynthesis.
d. There are three types of photosynthesis;
   (i) Name them and state the first product of Calvin cycle in each.
   (ii) Name any two plants or crops which use each of the photosynthesis types.
   (iii) Name the enzyme that fixes carbon dioxide.
   (iv) How do they types of photosynthesis differ among themselves?

READING BOOKS

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